Boost Converter Analysis and Design

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1 Boost Converter

1.1 Introduction

Boost Converter or Step Up Chopper is a type of DC-DC converter which increases the input DC voltage to a specified DC output voltage. It operates using energy storage elements like inductors and capacitors, along with switching components. The boost converter is widely used in applications requiring efficient voltage conversion, such as battery-powered systems and renewable energy systems.

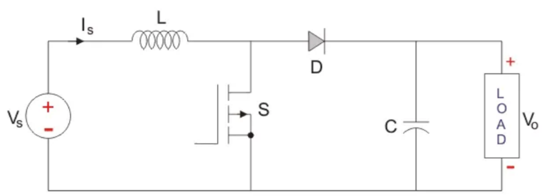

Figure 1: Schematic of boost converter

As shown in the figure, similar to the buck converter, the input voltage source is connected to a MOSFET, which acts as a switch. The second switch used is a diode. The diode is connected to a capacitor, and the load, and the two are connected in parallel.

1.2 Steady State Analysis

Similar to the buck converter, it is assumed here that the inductor current remains continuous, varying linearly as it rises and falls, while using the same terms and concepts.

1.2.1 Mode I: Switch is ON, Diode is OFF

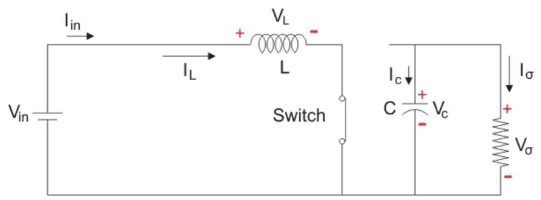

Figure 2: Equivalent circuit in Mode 1

In this mode, the switch is turned on, effectively acting as a short circuit with no resistance to the flow of current. As a result, all the current passes through the switch and returns to the DC input source.

Analyzing the circuit at steady state using KVL:

$$V_{\text{in}} = V_{L,\text{on}} \tag{1}$$

$$I_{C,\text{on}} = I_o \tag{2}$$

$$T_{\text{on}} = DT \tag{3}$$

1.2.2 Mode II: Switch is OFF, Diode is ON

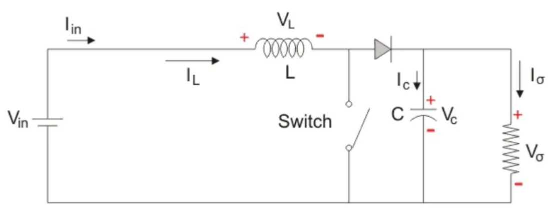

Figure 3: Equivalent circuit in Mode 2

In this mode, the polarity of the inductor reverses, releasing the energy it has stored during the previous phase. This energy is transferred to the load resistance, ensuring a continuous flow of current in the same direction and maintaining system stability. Additionally, the inductor works alongside the input source to boost the output voltage, effectively combining their potentials to achieve the desired step-up voltage.

Using KVL for analyzing the circuit in this mode:

$$V_{L,\text{off}} = V_{\text{in}} - V_o \tag{4}$$

$$I_{C,\text{off}} = I_L - I_o \tag{5}$$

$$T_{\text{off}} = (1 - D)T \tag{6}$$

Applying Volt-second balance:

$$V_{L,\text{on}}T_{\text{on}} + V_{L,\text{off}}T_{\text{off}} = 0 \tag{7}$$

Thus from equations (1), (3), (4), and (6):

$$V_o = \frac{V_{\text{in}}}{1 - D} \tag{8}$$

Applying Ampere-second balance:

$$I_{C,\text{on}}T_{\text{on}} + I_{C,\text{off}}T_{\text{off}} = 0 \tag{9}$$

Thus from equations (2), (3), (5), and (6):

$$I_L = \frac{I_o}{1 - D} \tag{10}$$

1.3 Design Analysis

1.3.1 Inductor Selection

The inductor value is chosen based on the desired current ripple ΔI_L:

$$L = \frac{V_{\text{in}}}{\Delta I_L} DT \tag{11}$$

or equivalently:

$$L = \frac{DV_{\text{in}}}{f\Delta I_L} \tag{12}$$

1.3.2 Capacitor Selection

The capacitor value is determined with respect to the capacitor voltage ripple ΔV_C:

$$C = \frac{DI_o}{f\Delta V_C} \tag{13}$$

2 MOSFET and its role

The Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) is a crucial switching device in power electronics. In the boost converter, the MOSFET acts as a high-speed switch, controlling the flow of current through the inductor. Its rapid switching capabilities ensure efficient energy transfer while minimizing losses. In this project, the MOSFET is driven by a pulse generator to achieve the desired switching frequency.

3 Simulation in MATLAB Simulink Using Simscape

The boost converter circuit was simulated using MATLAB Simulink with the Simscape plug-in. Each element in the model has a specific role:

- **DC Voltage Source**: Provides a fixed input voltage ($V_s = 12\,\text{V}$).
- **Inductor**: Stores and releases energy to facilitate voltage boosting ($L = 19.43\,\mu H$).
- **Capacitor**: Stabilizes and filters the output voltage ($C = 25\,\mu F$).
- **Resistor (Load)**: Represents the load consuming power ($R = 10\,\Omega$).
- **MOSFET and Pulse Generator**: The MOSFET acts as a switch, driven by a pulse generator with a duty cycle ($D = 0.5$) and switching frequency ($T = 1/40000\,\text{s}$).
- **Diode**: Ensures unidirectional current flow, directing energy to the load.

- **Voltage and Current Measurement Blocks**: These blocks continuously monitor input/output voltages and currents, providing critical data for analyzing circuit performance.
- **Scope**: It displays real-time waveforms of voltages and currents, enabling visual analysis and verification of the system's behavior during operation.
- **Connections**: Use of "Goto" and "From" blocks simplifies connections across the circuit.

4 Calculations

4.1 Output Voltage

A boost converter steps up the input voltage based on the duty cycle D:

$$V_o = \frac{V_{\text{in}}}{1-D}$$

With $V_{\text{in}} = 12\,\text{V}$ and $D = 0.5$:

$$V_o = \frac{12}{1-0.5} = \frac{12}{0.5} = 24\,\text{V}$$

The simulation result of 24.2 V is very close to this theoretical value, accounting for small losses (e.g., diode voltage drop, parasitic resistances). This confirms the circuit operates correctly at $D = 0.5$.

4.2 Inductor Current in Continuous Conduction Mode (CCM)

The boost converter is designed to operate in Continuous Conduction Mode (CCM) at $D = 0.5$, as stated in the document ("inductor current remains continuous"). The inductor current I_L in steady state is:

$$I_L = \frac{I_o}{1-D}$$

Where:

$$I_o = \frac{V_o}{R}$$

Given $V_o = 24\,\text{V}, R = 10\,\Omega$:

$$I_o = \frac{24}{10} = 2.4\,\text{A}$$

Substituting I_o into the equation for I_L:

$$I_L = \frac{2.4}{1-0.5} = \frac{2.4}{0.5} = 4.8\,\text{A}$$

4.3 Inductor Current Ripple

The inductor current ripple ΔI_L is given by:

$$\Delta I_L = \frac{V_{\text{in}} DT}{L}$$

Where:

$$T = \frac{1}{f} = \frac{1}{40000} = 25\,\mu\text{s},\ L = 19.43\,\mu\text{H}$$

Substituting the values:

$$\Delta I_L = \frac{12 \times 0.5 \times 25 \times 10^{-6}}{19.43 \times 10^{-6}}$$

$$\Delta I_L = \frac{12 \times 0.5 \times 25}{19.43} \approx 7.72\,\text{A}$$

4.4 Minimum and Maximum Inductor Currents

The inductor current oscillates between:

$$I_{L,\text{min}} = I_L - \frac{\Delta I_L}{2} = 4.8 - 3.86 = 0.94\,\text{A}$$

$$I_{L,\text{max}} = I_L + \frac{\Delta I_L}{2} = 4.8 + 3.86 = 8.66\,\text{A}$$

Since $I_{L,\text{min}} > 0$, the inductor current is continuous, confirming CCM operation at $D = 0.5$.

5 Conclusion

The design and simulation of the boost converter were successfully carried out in MATLAB Simulink using the Simscape toolbox. The simulation achieved an output voltage of 24.2V from a 12V input, demonstrating efficient voltage conversion. The analysis highlighted the importance of each circuit component in ensuring stable operation and efficient energy transfer. This work emphasizes the practical application of boost converters in modern electronics.

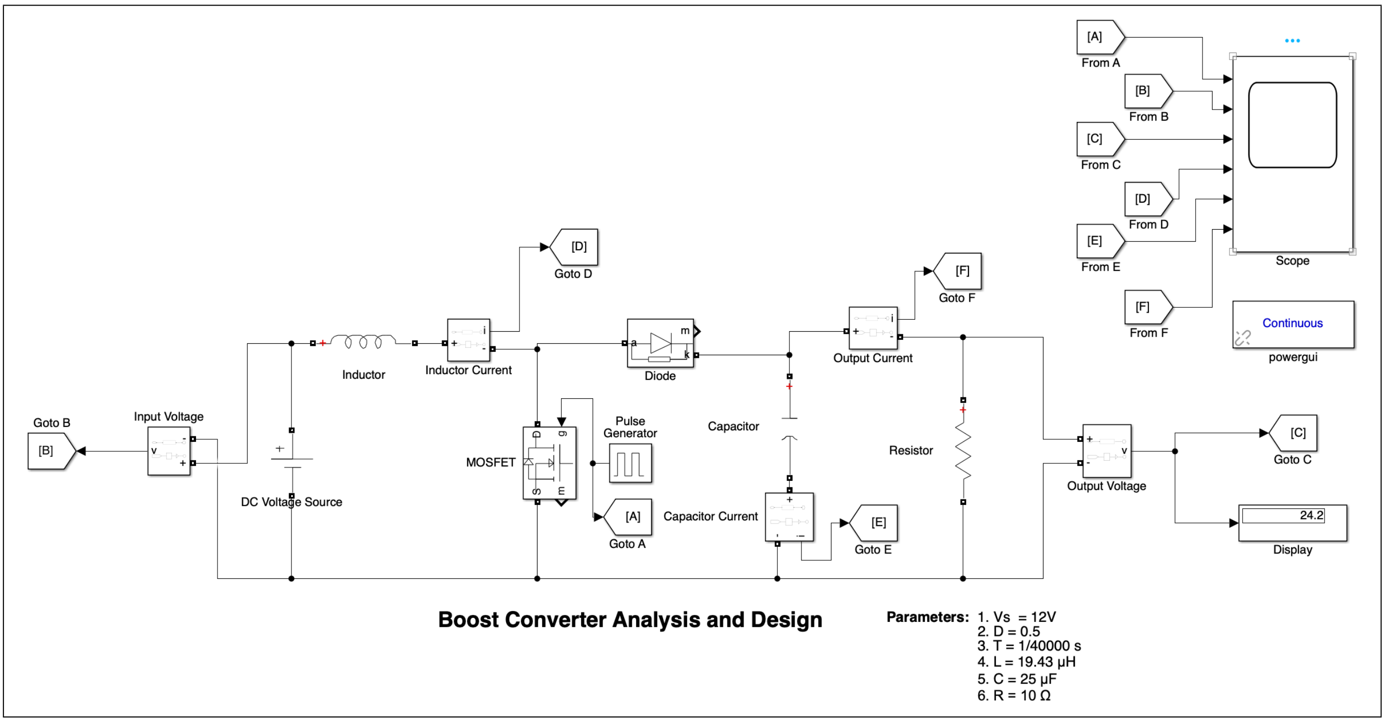

Figure 4: Simulation Model of Boost Converter

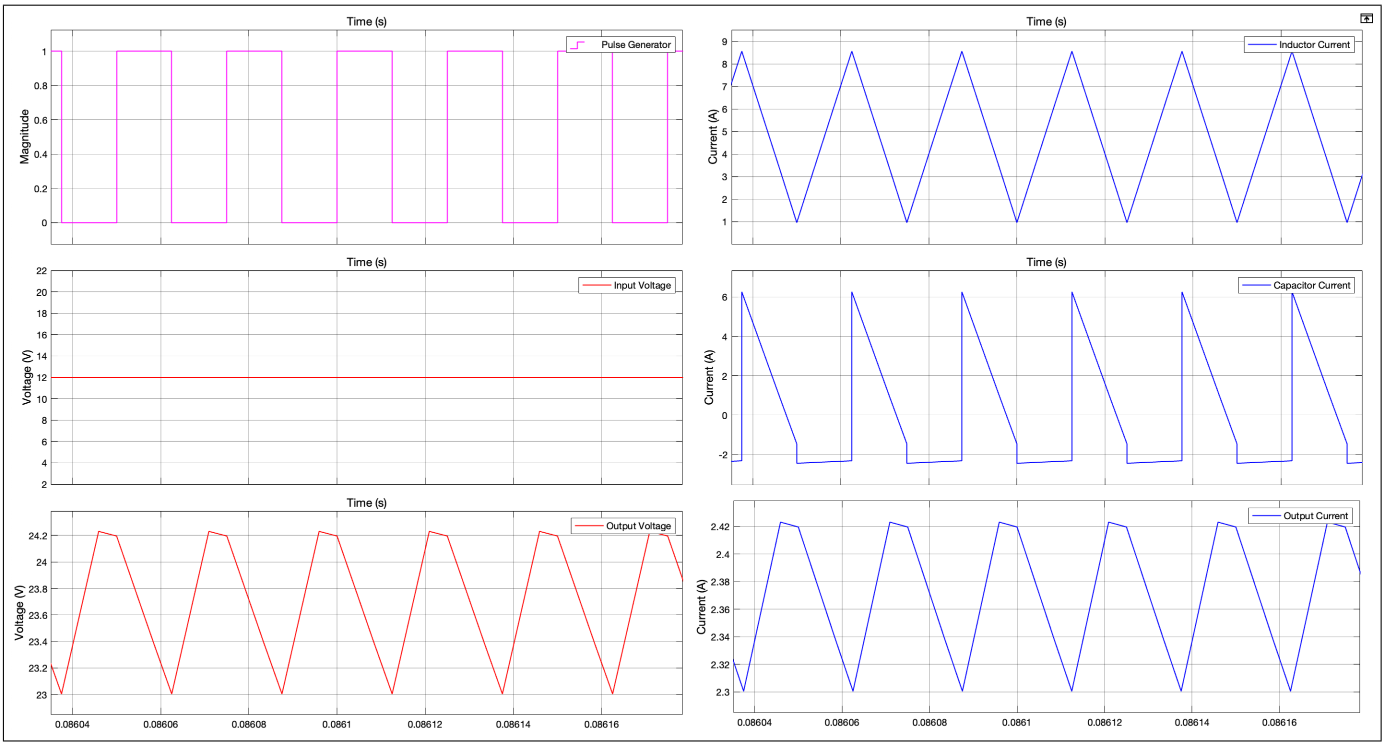

Figure 5: Waveforms of Boost Converter